Flexible Nets to optimize antibody production in chinese hamster ovary cells

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Problem

Antibodies are therapeutic proteins with a multitude of applications in **biotechnology** and medicine such as the treatment of viral infections, different types of cancer and common diseases such as psoriasis and multiple sclerosis.

The current design of **systems** for the production of antibodies is mainly based on **"trial and error"** methods that manipulate chinese hamster ovary cells (CHO cells). This leads to high expenditure of time and money, in addition to obtaining suboptimal process performance.

Contributions

The use of **mathematical models** has the potential to greatly accelerate and facilitate the design and optimization of **antibody production**.

Starting from a systematic and formal approach, the aim is to achieve an **automatic design** of the whole process that allows optimal productivity to be reached. To this end, we are developing both mathematical models and algorithms for the design and optimization of antibody manufacturing systems.

The mathematical models are based on:

Flexible Nets (FN), a novel modeling formalism able to accommodate uncertain parameters and nonlinear dynamics.

Method

FN consists of two connected subnets, an event net and an intensity net, each of which is a tripartite graph. The three vertices of these nets are:

- **Places** (◯): metabolites
- **Transitions** (▭): reactions
- **Handlers** (•): capture the way in which the occurrence of a reaction modifies the concentration of metabolites (event handler) and the way in which the concentration of metabolites modulates the reaction rates (intensity handler).

Uncertain parameters can be incorporated easily in an FN by associating inequalities to handlers.

References

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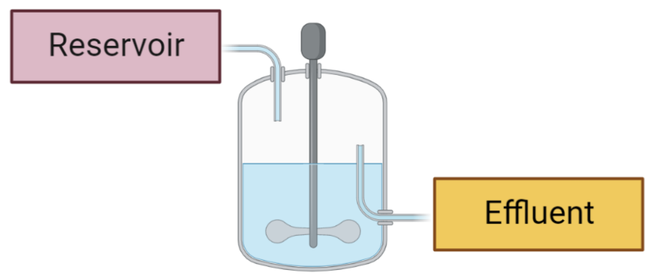
Case study

CHO cells are the most broadly used cells for the manufacturing of antibodies because of their advantages compared with other mammalian cells.

Flexible nets are used to develop a comprehensive model that encompasses both the **metabolic network** of CHO cells (iCHO1766) and the **dynamics** of the bioreactor, see Figure 1, in which the cells are cultured.

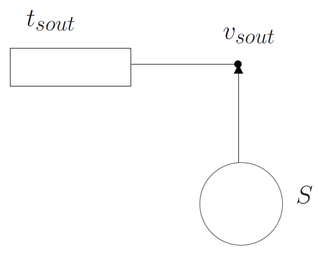


Thus, the model integrates **macroscopic** (dilution rate, substrate concentration, cell density, etc.) and **microscopic variables**, (intracellular metabolic fluxes, indicated as λ_i) simultaneously which enables a global optimization of the system.

Variables and parameters	Units	Description
X (macroscopic)	$gDWL^{-1}$	Cell density
S (macroscopic)	mM	Substrate or nutrient in the tank
D (macroscopic)	h^{-1}	Dilution rate
c (macroscopic)	mM	Nutrient concentration in medium
λ_i (microscopic)	$mmolgDW^{-1}h^{-1}$	Intracellular flux of reaction R_i in the cell

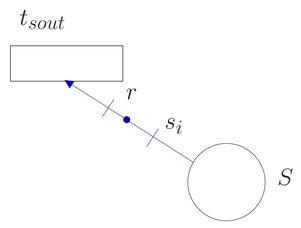
Modeling the stoichiometry

Event handler v_{gout} modeling the ouput flow of substrate from the tank.



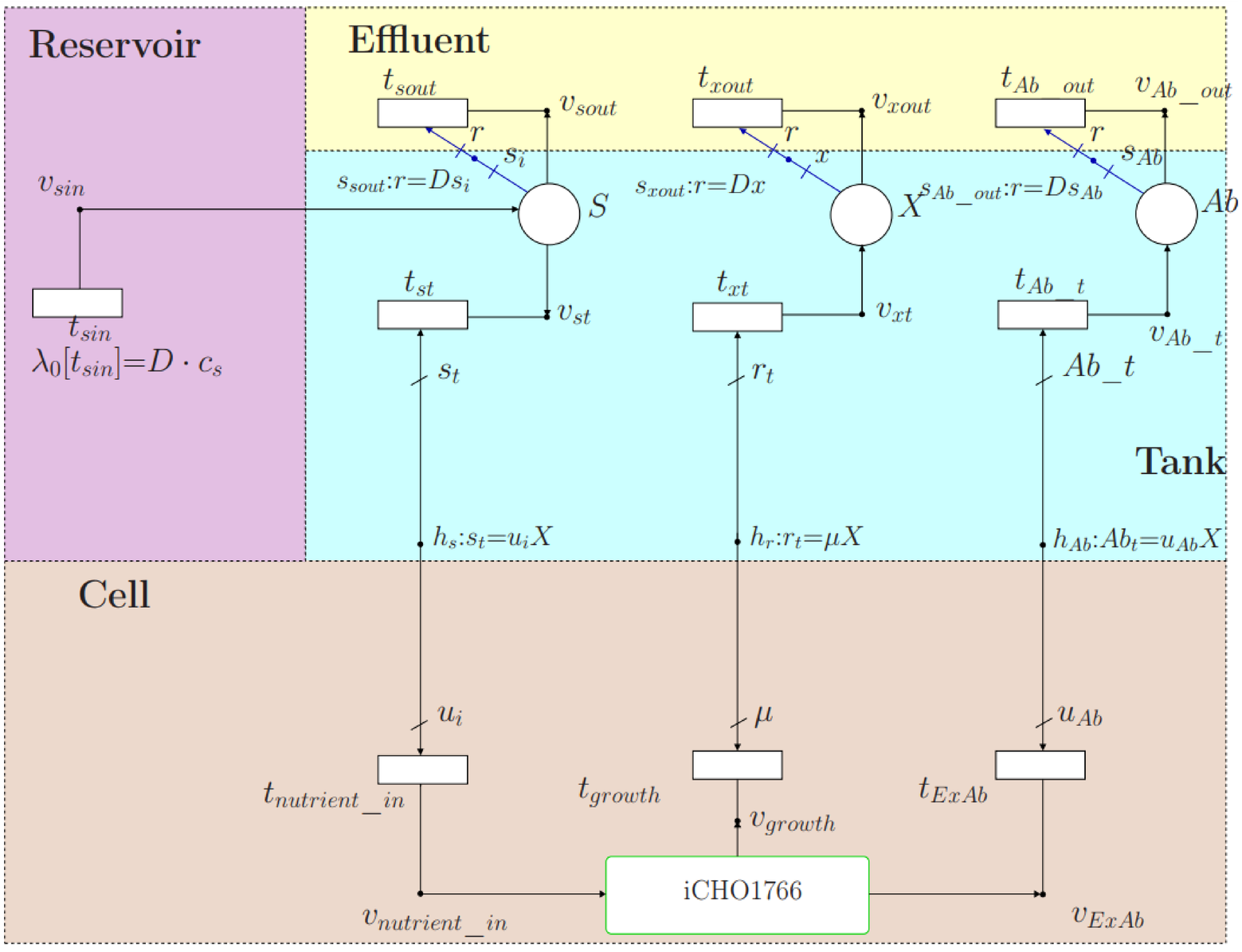
Modeling the kinetics

Intensity handler s_{gout} establishing that the output flow rate of substrate from the tank is $D \cdot S$.



Model Integration

FN integrating a metabolic network (Cell compartment) with the bioreactor dynamics (Reservoir, Tank and Effluent compartments) in continuous culture mode.



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